

IGNOU BCS-011 Solved Question Paper

Computer Basics and PC Software

Term-End Examination: June 2021

Time: 3 hours Maximum Marks: 100 Weightage: 75%

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Question 1

Question. (a) Convert the following numbers as asked: (i) $(12.5)_{10}$ to binary, (ii) $(1110\ 1000)_2$ to decimal, (iii) $(01111111000)_2$ to hexadecimal, (iv) $(BE5)_{16}$ to binary.

Answer.

(i) $(12.5)_{10}$ to binary: integer part $12 = 8 + 4$, so $12_{10} = 1100_2$. Fractional part: $0.5 \times 2 = 1.0$, so the fractional bit is 1. Hence

$$(12.5)_{10} = (1100.1)_2.$$

(ii) $(1110\ 1000)_2$ to decimal:

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 = 128 + 64 + 32 + 8 = 232.$$

Therefore, $(1110\ 1000)_2 = (232)_{10}$.

(iii) $(01111111000)_2$ to hexadecimal: group bits from the right into groups of four:

$$01111111000 \rightarrow 0011\ 1111\ 1000.$$

Now, $0011 = 3$, $1111 = F$, $1000 = 8$. Therefore,

$$(01111111000)_2 = (3F8)_{16}.$$

(iv) $(BE5)_{16}$ to binary: replace each hexadecimal digit by four bits:

$$B = 1011, \quad E = 1110, \quad 5 = 0101.$$

Hence,

$$(BE5)_{16} = (1011\ 1110\ 0101)_2.$$

Question. (b) Why does a computer need main memory? What is cache memory? How is main memory different from cache memory?

Answer. A computer needs main memory because the CPU can directly and quickly work only on data and instructions that are currently loaded in memory. Secondary storage such as hard disk or SSD is large but comparatively slower, so programs are first loaded into main memory before execution.

Main memory (RAM) stores the operating system, currently running programs, and data being processed. It is volatile, meaning its contents are lost when power is switched off.

Cache memory is a very small, very fast memory placed inside or close to the CPU. It stores frequently used instructions and data so that the CPU does not need to repeatedly fetch them from RAM.

Point	Main Memory/RAM	Cache Memory
Location	On motherboard as RAM modules	Inside or very near CPU
Speed	Fast	Faster than RAM
Capacity	Larger, usually GBs	Smaller, usually KBs or MBs
Cost per bit	Lower	Higher
Use	Holds active programs and data	Holds recently/frequently used data

Question. (c) What are the advantages and disadvantages of using inkjet printers?

Answer. Advantages:

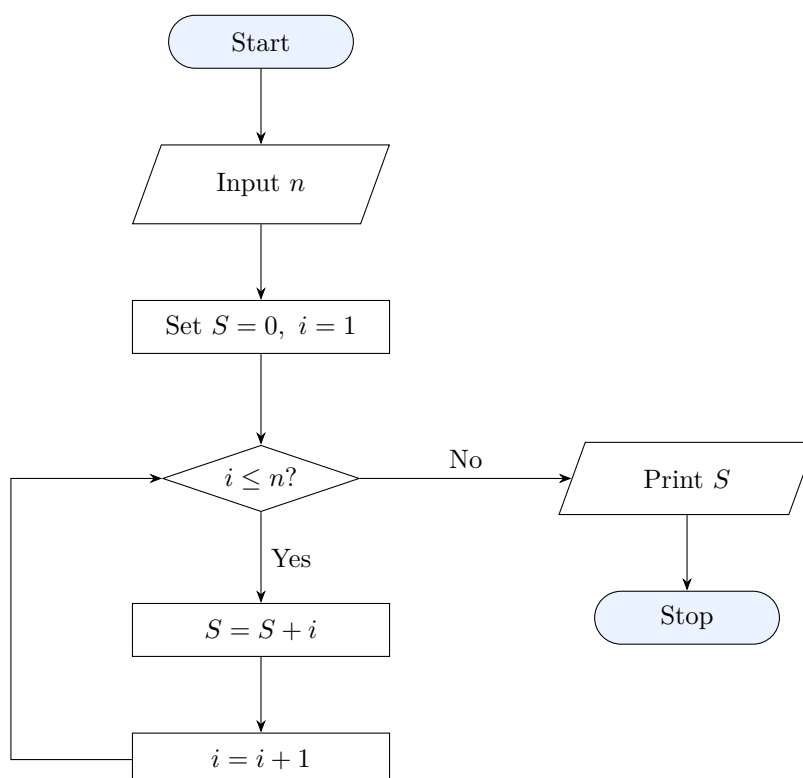
- They can print good quality colour images and photographs.
- They are comparatively cheap for home and small-office use.
- They are usually compact, quiet and easy to install.
- They can print on different types of paper such as photo paper, labels and envelopes.

Disadvantages:

- Ink cartridges may be costly in the long run.
- Print speed is generally lower than laser printers.
- Ink may dry if the printer is unused for a long time.
- Printed output may smudge if the ink or paper quality is poor.

Question. (d) Draw a flow chart to find the sum of the first n natural numbers.

Answer. The logic is to start with $i = 1$ and $S = 0$, repeatedly add i to S , and increase i until $i > n$. The final value of S is the required sum.



Question. (e) A University wants to maintain, manipulate and organise large student data. Which software will be needed by the University? Give justification in detail.

Answer. The University should use a **Database Management System (DBMS)** such as MySQL, PostgreSQL, Oracle, MS Access or a university ERP built on a DBMS.

A DBMS is suitable because student data is large, structured and regularly updated. It can store records such as enrolment number, name, programme, courses, marks, fees, attendance and

examination details.

Justification:

- **Organisation:** Data can be stored in related tables such as Student, Course, Fee, Marks and Attendance.
- **Fast searching:** Records can be searched quickly using keys such as enrolment number.
- **Manipulation:** Data can be inserted, updated, deleted and queried using SQL.
- **Integrity:** Rules can prevent duplicate enrolment numbers or invalid marks.
- **Security:** Different users can be given different permissions.
- **Reporting:** Mark sheets, fee reports, attendance reports and admission lists can be generated.

Question. (f) Explain the Internet-based architecture of software with the help of a diagram.

Answer. In Internet-based architecture, the user interacts with software through a browser or mobile app. The request travels through the Internet to a web server or application server. The server processes the request, communicates with a database server if needed, and sends the result back to the user.



Example: In an online admission system, the student submits a form through a browser; the application server validates it and stores it in the university database.

Question. (g) What are the different domains used in Web addresses? How is an IP address different from a Web address?

Answer. A web address or URL usually contains a domain name. Common domain categories are:

- **.com:** commercial organisations
- **.org:** organisations, often non-profit
- **.edu** or **.ac:** educational institutions
- **.gov:** government websites
- **.net:** network-related organisations
- **country domains:** .in for India, .uk for the United Kingdom, .us for the United States

Point	IP Address	Web Address/Domain Name
Form	Numeric, e.g. 192.168.1.1	Human-readable, e.g. www.ignou.ac.in
Purpose	Identifies a device/server on a network	Identifies a website in a memorable form
Used by	Network layer and routers	Users and browsers
Conversion	Directly used for routing	Converted to IP address by DNS

Question. (h) What is a Browser? What is a Cookie? Is cookie a threat to information security or privacy? Justify your answer.

Answer. A **browser** is application software used to access and display web pages. Examples are Google Chrome, Mozilla Firefox, Microsoft Edge and Safari.

A **cookie** is a small text file stored by a website in the user's browser. It may store login status, preferences, language choice, shopping-cart details or tracking identifiers.

A cookie is not always harmful. Many cookies are useful because they remember user preferences and maintain sessions. However, cookies can become a **privacy risk** when third-party advertisers or websites use them to track browsing behaviour across sites. They may also be a security risk if session cookies are stolen. Therefore, users should block unnecessary third-party cookies, clear cookies regularly and use HTTPS websites.

Question. (i) List and explain any two processes that are part of e-learning systems.

Answer. Two important processes in an e-learning system are:

1. **Content delivery:** The system provides digital learning material such as notes, videos, presentations, simulations and reading resources through a learning management system.
2. **Assessment and feedback:** Students attempt quizzes, assignments or online tests. The system evaluates answers automatically or allows teachers to grade them and provide feedback.

Other processes may include discussion forums, attendance tracking, course registration, video conferencing and progress monitoring.

Question 2

Question. (a) Identify the Net ID and Device ID for the following IP address and subnet mask pairs: 192.168.5.97 and 255.255.255.0; 10.10.13.10 and 255.255.0.0.

Answer. The Net ID is obtained by performing a bitwise AND between the IP address and subnet mask. The device or host ID is the remaining part.

IP Address	Subnet Mask	Net ID	Device/Host ID
192.168.5.97	255.255.255.0	192.168.5.0	0.0.0.97
10.10.13.10	255.255.0.0	10.10.0.0	0.0.13.10

Question. (b) How does the search on keyword “Intelligent” OR “System” differ from the search “Intelligent” AND “System”? Explain.

Answer. **OR search** gives broader results. The query **Intelligent OR System** retrieves pages containing either the word “Intelligent”, or the word “System”, or both. It increases the number of results.

AND search gives narrower results. The query **Intelligent AND System** retrieves only those pages where both words are present. It reduces irrelevant results and is more specific.

Example: a page containing only “Intelligent machines” may appear in the OR search but not in the AND search. A page containing “Intelligent System design” will appear in both searches.

Question. (c) What is a MODEM? Why is it used in networking? Explain.

Answer. A **MODEM** means **modulator-demodulator**. It converts digital signals from a computer into a form suitable for transmission over a communication medium, and converts incoming transmitted signals back into digital form.

It is used in networking because computers generate digital data, while many communication lines may require analog or carrier-based signals. For example, in broadband or telephone-line communication, a modem helps a computer or router connect to the Internet service provider.

Functions of a modem:

- Converts signals for transmission and reception.
- Provides Internet connectivity through ISP lines.
- May include routing, Wi-Fi and firewall features in modern home gateways.
- Helps multiple devices access the Internet when combined with a router.

Question. (d) Explain the characteristics and advantages of Wide Area Network (WAN).

Answer. A **Wide Area Network (WAN)** connects computers and networks over a large geographical area such as cities, states, countries or continents. The Internet is the largest example of a WAN.

Characteristics:

- Covers large geographical distance.
- Connects multiple LANs and MANs.
- Uses public or leased communication lines, satellite links, fibre links or telecom networks.
- Usually requires routers and other network devices.
- Data speed may be lower than LAN due to distance and shared infrastructure.

Advantages:

- Allows branches of an organisation to communicate.
- Enables sharing of central databases and applications.
- Supports e-mail, video conferencing, cloud services and remote work.
- Provides global connectivity and access to Internet resources.

Question. (e) What are the characteristics of twisted pair cable? Compare its features with optical fibre cables.

Answer. Twisted pair cable consists of pairs of insulated copper wires twisted together. Twisting reduces electromagnetic interference and crosstalk.

Characteristics of twisted pair cable:

- Low cost and easy to install.
- Commonly used in telephone lines and Ethernet LANs.
- Available as UTP (unshielded twisted pair) and STP (shielded twisted pair).
- Suitable for short to medium distances.
- More affected by noise than optical fibre.

Point	Twisted Pair Cable	Optical Fibre Cable
Medium	Copper wire	Glass/plastic fibre
Signal	Electrical signal	Light signal
Bandwidth	Lower	Very high
Distance	Shorter	Longer
Noise immunity	Lower	Very high
Cost	Cheaper	Costlier
Installation	Easier	Requires more care

Question 3

Question. (a) Explain how object-oriented software design paradigm is different from structured and modular design paradigm.

Answer.

Point	Object-Oriented Design	Structured Design	Modular Design
Basic unit	Object/class	Function/procedure	Module
Data and operations	Data and methods are combined	Data and functions are separate	Related functions are grouped in modules
Main idea	Encapsulation, inheritance, polymorphism	Top-down decomposition of tasks	Divide software into manageable units
Reusability	High due to classes and inheritance	Moderate	Moderate to high
Example	Student object with data and methods	Separate functions for input, process, output	Separate admission, fee and exam modules

Object-oriented design is closer to real-world modelling because it represents entities as objects. Structured design focuses on the sequence of operations, while modular design focuses on dividing the system into independent modules.

Question. (b) What is the need of an Operating System? Give names of two popular operating systems. Suppose you want to transfer a file from your hard disk to a CD, which OS services will be used and for what purpose?

Answer. An **Operating System (OS)** is needed because it acts as an interface between user, application software and hardware. It manages CPU, memory, files, input/output devices and security. Examples: **Microsoft Windows** and **Linux**.

For transferring a file from hard disk to CD, the following OS services are used:

OS Service		Purpose in file transfer from hard disk to CD
User Interface		Allows the user to select the file and choose copy/burn command.
File Management		Locates, reads and organises files and folders on disk.
Input/Output Services		Controls hard disk reading and CD writer output operation.
Process Control		Manages the running copy/burning process and prevents conflicts.
Memory Management		Allocates buffer memory for temporary storage during transfer.

Question. (c) Explain which software will be required to perform the following tasks. Also list four features of these software: (i) scheduling of tasks, (ii) creating mail merge, (iii) entering formulas and functions, (iv) creating presentation.

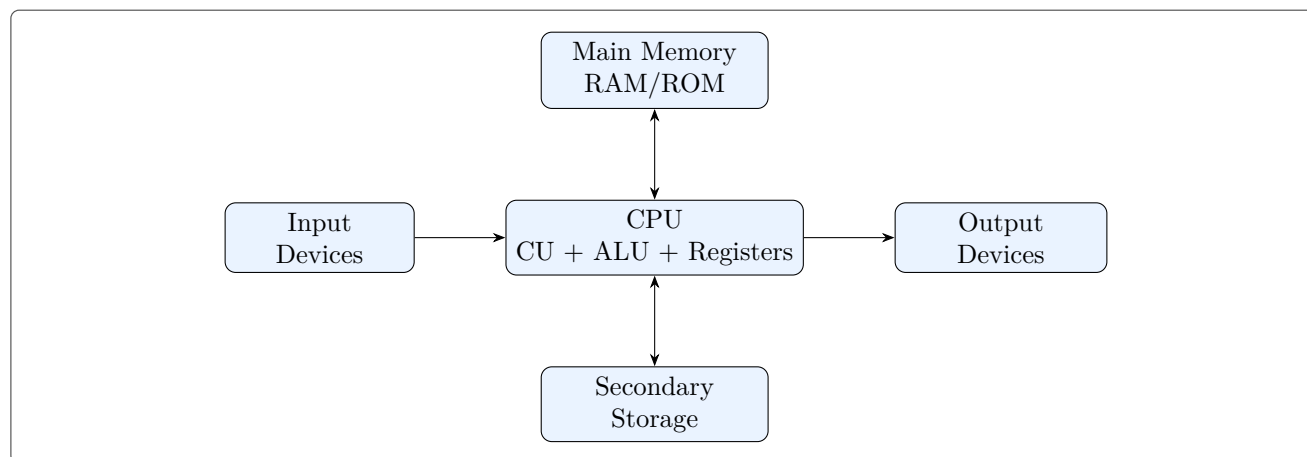
Answer.

Task		Required Software	Four important features
Scheduling of tasks		Calendar/task management or project management software	Due dates, reminders, priority setting, recurring tasks/dependencies
Creating mail merge		Word processor	Main document, data source connection, merge fields, printing/e-mailing personalised documents
Entering formulas and functions		Spreadsheet software	Cells and ranges, formulas, built-in functions, charts/tables
Creating presentation		Presentation software	Slides, templates/themes, animations/transitions, insertion of images/audio/video

Question 4

Question. (a) Explain the working of a basic computer with the help of a diagram showing its basic components.

Answer. A basic computer works on the input-process-output principle. Input devices enter data and instructions. The CPU processes them using the control unit, ALU and registers. Main memory stores active instructions and data. Output devices display or print the result. Secondary storage stores data permanently.



Question. (b) What are the differences between primary and secondary memory of a computer? Which of them is larger in size? Which of them is faster?

Answer.

Point	Primary Memory	Secondary Memory
Examples	RAM, ROM, cache	Hard disk, SSD, CD/DVD, pen drive
CPU access	Directly accessed by CPU	Accessed through I/O operations
Speed	Faster	Slower than primary memory
Size	Smaller	Larger
Volatility	RAM is volatile; ROM is non-volatile	Usually non-volatile
Use	Holds current programs/data	Permanent storage of files and software

Secondary memory is usually **larger in size**. Primary memory is usually **faster**.

Question. (c) Explain the role of the following utility software: (i) Disk checkers, (ii) System restore.

Answer.

- (i) **Disk checkers:** A disk checker scans storage media for file-system errors, bad sectors and lost clusters. It helps repair logical errors and improves reliability. Example: Check Disk in Windows.
- (ii) **System restore:** System restore returns system files, settings and registry entries to an earlier restore point. It is useful when a driver installation, update or software change creates problems. It usually does not delete personal documents.

Question. (d) What is the use of digitizing (graphic) tablet and joystick?

Answer. A **digitizing or graphic tablet** is an input device used to draw, trace or write directly on a flat surface using a stylus. It is useful in digital art, CAD, map digitisation, signature capture and handwritten input.

A **joystick** is an input device used to control movement and direction. It is commonly used in games, flight simulators, robotics control and some industrial applications.

Question 5

Question. Explain any five of the following with the help of an example/a diagram: (a) Supercomputer, (b) Fourth Generation Computers, (c) Disk Layout of a CD-ROM, (d) Perverse Software, (e) Open-source Software, (f) Wiki, (g) TCP/IP.

Answer. The question asks any five; answers for all seven are given for complete preparation.

(a) Supercomputer

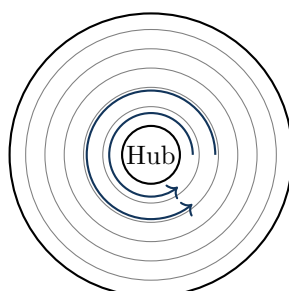
A supercomputer is an extremely powerful computer designed to perform very large numbers of calculations per second. It uses parallel processing and many processors working together. Supercomputers are used for weather forecasting, climate modelling, nuclear simulations, space research, cryptography and large scientific computations. Example: PARAM series supercomputers in India.

(b) Fourth Generation Computers

Fourth generation computers are based on microprocessors using Very Large Scale Integration (VLSI) technology. They are smaller, faster, cheaper and more reliable than earlier generations. Personal computers, laptops and many modern embedded systems belong to this generation.

(c) Disk Layout of a CD-ROM

A CD-ROM stores data optically in the form of pits and lands on a spiral track. A laser beam reads the reflected light from the surface.



Data is arranged along a spiral optical track

(d) Perverse Software

Perverse software is harmful software that intentionally damages, disrupts or misuses a computer system. It includes viruses, worms, Trojan horses, spyware, ransomware and logic bombs. Example: ransomware encrypts user files and demands payment for decryption.

(e) Open-source Software

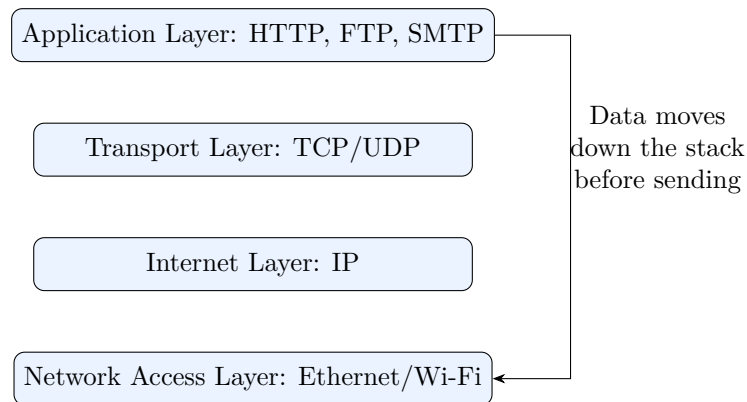
Open-source software is software whose source code is available to users for study, modification and redistribution under an open-source licence. Examples include Linux, LibreOffice, Firefox and GIMP. Its advantages include low cost, community development, transparency and freedom to customise.

(f) Wiki

A wiki is a collaborative web application where users can create, edit and link pages using a browser. It is useful for knowledge sharing, documentation, class notes, project collaboration and FAQs. Example: Wikipedia.

(g) TCP/IP

TCP/IP is the main protocol suite used on the Internet. IP is responsible for addressing and routing packets from source to destination. TCP provides reliable delivery by breaking data into segments, numbering them, checking errors and retransmitting lost data.

**Quick Revision Table**

Topic	Key Point
Main memory	Active working area for CPU; RAM is volatile.
Cache	Very fast small memory close to CPU.
DBMS	Best for large organised student records.
WAN	Connects networks over large geographical area.
Subnet mask	Separates Net ID and Host ID.
OS	Manages hardware, software, files, memory, I/O and processes.
TCP/IP	Protocol suite for Internet communication.